have we no reason for viewing them with a certain mistrust? For if atoms are supposed to be the smallest units of matter, they cannot be expected to behave in every respect like the visually perceptible objects of our daily experience—like the actual black and white dots in the spatial model of a crystal, for instance. One is indeed quite justified in suspecting that here, in a region where we are approaching the ultimate, fundamental component parts of matter, there is a limit to our power of perception, too. Hence, we must ask, first of all, what the actual size of the atoms is, and what magnification would be required to make a molecule appear to us the size, say, of a billiard ball. Secondly, we must ask, to what extent such a visual model can be justified at all—in other words, what its intended purpose is. Does it possess such a directly perceptible import that we may expect an ideally perfect microscope of the future actually to show us such an image of a real molecule?

Let us take first the question of the size of atoms. Obviously, not all atoms are the same size, but all atoms are more or less of the same order of magnitude. The magnification necessary to permit us to see an atom as a structure with a diameter of, roughly, 10 centimetres, is approximately that which we would have to use to magnify a ball of 1 centimetre in diameter to the size of the earth. This example will give you an approximate idea of the infinitesimal smallness of molecules.

Now, the second question, that of the significance of a molecular model: In recent years, a quite novel microscope has been developed, the *electron microscope,* which—unlike the ordinary optical microscope—does not utilize light rays, but electron rays. With this electron microscope, a considerably greater resolving power, and a far greater magnification, can be attained than with the optical microscope, so that it enables us, even at the present stage of its development, to see particularly large molecules as discrete particles. If it should ever become possible to increase this magnifying power to twenty or thirty times its present degree—to be sure, a difficult problem—an individual water molecule may conceivably be made visible through such a microscope.

But the important question is whether we would then see anything in any way resembling the model shown in Figure 1. It is true, of course, that no molecule is ever in a state of rest. It moves, as a whole, under the influence of the temperature, and its component parts vibrate, in reciprocal movement. Thus it would be necessary to take cinematographic records of molecules, in which case one would actually obtain a snapshot of the kind shown in Figure 1. This cannot be doubted in the light of all our present knowledge of atomic physics—and this realization is at the same time a recognition of the visual-perceptual significance of models of the kind shown in Figure 1. However, as a result of thermal movement, minor changes would be constantly taking place in this picture.

How does the physicist know that a water molecule consists just exactly of